

Maci Kesler

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Center for Nuclear Research

Department of Physics

Kent State University

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Education

- 2022 – Present  **Ph.D., Physics, Kent State University**
Advisor: Zhangbu Xu
Research Interests: diffractive electron-nucleus scattering, heavy-ion/electron-ion collisions, nuclear structure, quark-gluon plasma, vector meson production
- 2022 – 2024  **M.A., Physics, Kent State University**
- 2019 – 2022  **B.S., Computational Physics, University of Nevada, Las Vegas**
Summa Cum Laude
Honors Thesis: *Dynamics of magnetic monopoles with charged matter*
- 2018–2019  **A.A., Elementary Education, College of Southern Nevada**

Appointments

- 2024 – Present  **Graduate Research Assistant**, Kent State University, Kent, OH, USA
- 2022 – 2024  **Graduate Teaching Assistant**, Kent State University, Kent, OH, USA

Service

- 2026 – 2027  **City Coordinator - Pint of Science**, Northwater Brewing, Kent, OH, USA
- 2025 – 2026  **President - Physics Graduate Student Association**, Kent State University, Kent, OH, USA
-  **Event Manager - Pint of Science**, Northwater Brewing, Kent, OH, USA
- 2025  **SciPals Outreach Program**, Miller South School for the Visual and Performing Arts, Akron, OH, USA
-  **STAR Collaboration Meeting Organizer**, Kent State University, Kent, OH, USA
-  **Fascinating Physics Tour Guide**, Brookhaven National Lab, Upton, NY, USA
-  **Host - Physics Department Flash Talk Series**, Kent State University, Kent, OH, USA
-  **Nuclear Physics Day on the Hill**, Washington, D.C., USA
-  **Introduction to Particle Physics Lecture**, Kent State University, Kent, OH, USA
- 2024 – 2025  **Organizer - Center for Nuclear Research**, Kent State University, Kent, OH, USA
- 2024  **STAR Shifts**, Brookhaven National Lab, Upton, NY, USA
-  **Student Representative - Student Academic Complaint Committee**, Kent State University, Kent, OH, USA
-  **Event Organizer - Total Solar Eclipse**, Kent State University, Kent, OH, USA
- 2023 – 2025  **Graduate Program and Admission Committee Representative - Physics Graduate Student Association**, Kent State University, Kent, OH, USA
- 2023  **Development of Quantum Computing Training Program**, Kent State University, Kent, OH, USA
- 2020 – 2022  **President - Society of Physics Students**, University of Nevada, Las Vegas, NV, USA

Computing

- **Programming Languages:** C++, Mathematica, Python, Qiskit, ROOT, PyTorch
- **Software:** Linux, Git/GitHub, Jupyter Notebook, VS Code, HTCondor, Rucio

Achievements

Awards

- 2025 ■ **Department of Energy - Office of Science: Graduate Student Research Award**, Brookhaven National Lab, Upton, NY, USA
- 2024 ■ **Departmental Graduate Award: Outstanding Service Award**, Kent State University, Kent, OH, USA
- **Department of Physics Scholarship**, Kent State University, Kent, OH, USA
- 2022 – 2024 ■ **Graduate Dean's Scholarship/Award**, Kent State University, Kent, OH, USA
- 2020 – 2021 ■ **Frank Honors Scholarship**, University of Nevada, Las Vegas, NV, USA
- **Russell L. Brenda Frank Endowed Scholarship**, University of Nevada, Las Vegas, NV, USA
- 2020 ■ **NASA Nevada Space Grant**, University of Nevada, Las Vegas, NV, USA

Certifications/Summer Schools

- 2026 ■ **BNL-CERN School on Physics at Future Colliders - Expanding our Horizons**, European Organization for Nuclear Research, Geneva, CHE
- 2024 ■ **Center for Frontiers in Nuclear Science Summer School - Physics of the Electron-Ion Collider**, Stony Brook University, Stony Brook, NY, USA
- 2023 ■ **Qiskit Global Summer School 2023 - Quantum Excellence, Theory to Implementation**, Online
- 2020 ■ **Graduate Research Certification - Grad Rebel Advantage Program**, University of Nevada, Las Vegas, NV, USA

Collaborations/Working Groups

- 2026 – Present ■ **H1**, DESY (Deutsches Elektronen-Synchrotron) Lab, Hamburg, DEU
- 2025 – Present ■ **ePIC**, Brookhaven National Lab, Upton, NY, USA
- **Exclusive/Diffractive/Tagging** Physics Working Group for ePIC, Brookhaven National Lab, Upton, NY, USA
- 2024 – Present ■ **STAR**, Brookhaven National Lab, Upton, NY, USA
- **EIC Users Group**, Brookhaven National Lab, Upton, NY, USA

Conference Presentations

- 2026 ■ **Towards Improved Hadron Femtography, Edition V**, "Analysis and Reconstruction Strategies for Exclusive Diffractive eA Physics at ePIC", University of Virginia, Charlottesville, VA, USA
- **Nuclear Physics Seminar**, "Imaging Nuclei Through Exclusive Vector Meson Production", Brookhaven National Lab, Upton, NY, USA
- **ePIC and EIC Physics Readiness Workshop**, "Incoherent Background Suppression in ϕ Production in $e+A$ Collisions", University of Calabria, Cosenza, ITA
- **ePIC Collaboration Meeting**, "Diffractive ϕ in eAU Collisions at ePIC", Brookhaven National Lab, Upton, NY, USA

Conference Presentations (continued)

- 2025
- **ePIC and EIC Physics Readiness Workshop**, "Reimagining $|t|$: Advanced Approach to Momentum Transfer Measurement", Institute of Physics, London, GBR
 - **Center for Nuclear Research Seminar**, "From Diffraction to Definition: Imaging Nuclei with Vector Mesons", Kent State University, Kent, OH, USA
 - **EIC/ePIC Joint Collaboration Meeting**, "Nuclear Imaging Through Coherent Exclusive Vector Meson Production", Jefferson Lab, Newport News, VA, USA
 - **EIC Early Science Workshop**, "Improving $|t|$ Measurement Through Exclusive Coherent Vector Meson Production", Stony Brook University, Stony Brook, NY, USA
 - **Graduate Student Symposium**, "Projective Imaging of Nuclei at the Electron-Ion Collider", Kent State University, Kent, OH, USA
 - **Physics Opportunities at an Electron-Ion Collider**, "Projective Imaging of High-Energy Nuclei Through Exclusive Vector Meson Production in Electron-Ion Collisions", Florida International University, Miami, FL, USA
- 2024
- **Center for Nuclear Research Seminar**, "On Diffractive Physics", Kent State University, Kent, OH, USA
 - **American Physical Society - Eastern Great Lakes Section**, "Investigating Novel Ways of Improving Nuclear Imaging Through Exclusive Vector Meson Production at the Electron-Ion Collider", Marietta University, Marietta, OH, USA